

Reference

Supporting detail for the case study: code evidence, full technology stack, cost inputs, KPI trajectory and glossary. Code facts come from the public [Apskaita5 repository](#) at commit `2320f1c`.

Codebase at a glance

Layer	Project	.vb files	Lines	What it holds
Business objects	ApskaitaObjects	537	304,727 (about 222,000 hand-written)	377 CSLA classes, 425 data-portal methods, 797 rule registrations, 28 declaration classes
Desktop forms	AccDataBindingsWinForms	332	147,860 (101,658 designer-generated)	129 of the 156 forms; 114 editable grids with 1,846 columns
Controls	AccControlsWinForms	82	32,624	Keyboard-friendly pickers, numeric and date boxes, report printing
Data access	AccDataAccessLayer	68	16,625	Named-SQL dictionary, transactions, schema compare and fix, users and roles
Shell, reports, templates	Apskaita (exe)	34	11,939	Main window, 68 RDLC reports, 24 FFDData and 24 MXFD statutory templates

Layer	Project	.vb files	Lines	What it holds
Servers, plugins, helpers	Remoting server, web service, plugin manager, helpers, migration tool, invoice format	56	8,196	Data portal over HTTP, a small REST endpoint for invoices, an invoice-import adapter interface
Total		1,109	521,971	

Code evidence

Paths are relative to `Source/`.

Fact	Where to look
Schema: 67 tables, 829 columns, DOUBLE money, no foreign keys	<code>Apskaita5/Apskaita5/DbStructure/DatabaseStructureGauge.xml</code>
VB rounding helper (compares through Decimal)	<code>Apskaita5/AccCommon/CommonMethods.vb</code> , lines 14 to 24
MySQL <code>CROUND</code> stored function (compares in DOUBLE)	<code>DatabaseStructureGauge.xml</code> , stored procedure list
Named SQL dictionary, 770 MySQL and 761 SQLite statements	<code>AccDataAccessLayer/AccDataAccessLayer/SqlDepositories/*.sqld</code>
Schema compare and fix; old-version sniffing	<code>AccDataAccessLayer/AccDataAccessLayer/DatabaseAccess/DatabaseStructure/DatabaseStructureErrorList.vb</code> ; <code>Apskaita5/Apskaita5/CustomDatabaseStructure/ErrorMessageManager.vb</code>
One class per declaration version, hardcoded validity dates	<code>Apskaita5/Apskaita5/ActiveReports/Declarations/</code> (for example <code>DeclarationSAM_7.vb</code>)
Declaration list hardcoded in the forms layer	<code>Apskaita5/AccDataBindingsWinForms/CommonMethods.vb</code> , lines 577 to 621
Hardcoded 2019 social-insurance gross-up of 1.289	<code>Apskaita5/Apskaita5/ActiveReports/WageVDUInfo.vb</code> , line 151
Per-company editable non-taxable income formula	<code>Apskaita5/Apskaita5/General/CompanyObjects/Company.vb</code> , line 49

Fact	Where to look
Every app user can be a MySQL user (grants issued by the app)	<code>AccDataAccessLayer/.../MySQLGenerator.vb</code> , lines 814 to 1068
Global company context, 321 uses	<code>GetCurrentCompany()</code> in <code>Apskaita5/Apskaita5/Utilities.vb</code>
Grid keyboard behavior (Insert, Delete, Enter, Tab)	<code>Apskaita5/AccDataBindingsWinForms/DataListViewEditControlManager.vb</code> , lines 1260 to 1300
Date shorthand entry	<code>Apskaita5/AccControlsWinForms/Controls/AccDatePicker.vb</code> , lines 310 to 340
Positional report data set, 345 columns	<code>Apskaita5/AccControlsWinForms/ReportData.xsd</code>
Plaintext database credential	<code>Apskaita5/AccWebService/Web.config</code> , line 58

The rounding disagreement was found by reading both implementations and emulating them over every half-cent value from 0.005 to 9,999.995: 6.6 percent round differently, as do 354 of 200,000 random 21 percent VAT amounts. The L3 harness confirms this on the real runtimes in weeks 2 to 4.

Technology stack

Layer	Choice
Web client	React, TypeScript, AG Grid, installable PWA; Playwright keyboard tests
API and modules	.NET 10 (LTS), ASP.NET Core, C#; EF Core with Npgsql, Dapper for heavy reads
Database	Azure Database for PostgreSQL Flexible Server, zone-redundant, one read replica per cell
Compute	Azure Container Apps per cell; migration factory as Container Apps jobs
Messaging	Azure Service Bus with a transactional outbox
Edge and identity	Azure Front Door with WAF; Microsoft Entra External ID (OIDC, MFA)
Keys and secrets	Azure Key Vault, managed identities, per-tenant data keys

Layer	Choice
Documents	Azure Blob Storage with immutability policies
AI services	Azure AI Document Intelligence; Azure OpenAI (EU Data Zone) behind a model gateway; pgvector
Observability	OpenTelemetry to Azure Monitor and Grafana
Delivery	GitHub Actions, Terraform, feature flags, canary cell first
Modernization tooling	Roslyn analysers for VB.NET and C#; code models through the gateway; FsCheck property tests; mutation testing; legacy harness in Windows containers on an AKS Windows node pool

AWS equivalents work as well: ECS Fargate, RDS PostgreSQL, SQS, S3, KMS.

Non-functional targets

Area	Target
Availability	99.9% a month; 99.95% from the 15th to the 25th, when filings fall due
Recovery inside a region	Zone-redundant databases: no data loss, service back within 1 hour
Disaster recovery	Standby copy in a second EU region: at most 15 minutes of data lost, service back within 4 hours
Peak load	Each cell sized for 3 times its average load in filing week
Speed	Keystroke to screen under 50 ms; API under 300 ms at the 95th percentile; month-end reports under 10 seconds
Data residency	All customer data and AI processing in EU regions

Cloud cost inputs

Planning estimates from list prices, validated against a real bill in week 4.

Item	Unit cost	Quantity at full scale
Cell database: PostgreSQL, 16 vCores, zone-redundant, with a read replica	3,500 EUR a month	7 cells
Cell compute: application containers and workers	2,000 EUR a month	7 cells
Cell storage, backup and network	800 EUR a month	7 cells
Dedicated database	150 EUR a month	about 126 tenants
Shared: edge, identity, observability, control plane, non-production	25,000 EUR a month	1
AI inference and document extraction	4 EUR per tenant a month	4,200 tenants
Total		1.26M EUR a year, or 299 EUR per customer

Investment: a delivery team of 32 for 12 months and 10 for six more at a blended 7,300 EUR a month (3.24M EUR), modernization tooling and AI usage (0.25M EUR), and 15 percent contingency on year 1 (0.46M EUR). The firm's own staff on the program are existing payroll and are not counted. The model is `case-study/cost_model.py` in the repository.

KPI trajectory

KPI	Month 6	Month 9	Month 12
Modules migrated (of 9)	5	9	9, with all extension tiers
Differential test pass rate	100% on W0 modules	100% on released modules	100%; no unsigned deviations
AI code accepted versus reworked	Baseline on Ledger	Tracked per module	Tracked per module; no quota
Customers migrated (stable 5 business days)	80	1,170	2,970

KPI	Month 6	Month 9	Month 12
Cloud run cost per customer	Not meaningful under 100 tenants	490 EUR or less	406 EUR or less (model: 330 EUR)
Statutory turnaround	First pack change within 20 days	Every change within 20 days	Every change within 20 days; median 14
Realized savings (annual run rate)	-0.38M EUR	-0.38M EUR	+0.03M EUR (break-even); 1.84M EUR from month 21

Glossary

Term	Meaning
VMI	Lithuanian State Tax Inspectorate
Sodra	Lithuanian State Social Insurance Fund
GPM, PSD, NPD	Personal income tax; compulsory health insurance; non-taxable income amount
FR0600, i.SAF, SAF-T	VAT return; monthly VAT invoice register; standard audit file for tax
SAM, 13-SD, GPM313	Sodra monthly and event declarations; monthly income tax declaration
FFData, MXFD	Data and form-definition files of the e-form software used for Lithuanian declarations
camt.052, camt.053, pain.001	ISO 20022 bank statement and payment messages
CSLA	.NET business-object framework used by the legacy code
Row-level security (RLS)	Database rule that filters every row by tenant for every query
Cell	Self-contained slice of the platform serving about 600 tenants
Transactional outbox	Events written in the same transaction as the data, so none are lost; consumers ignore repeats
Authority stamp	Version number marking which system may write a tenant's books; stale commands are rejected

Term	Meaning
Semantic layer	Catalog of business metrics with fixed definitions that queries must use
Calibrated confidence	A model score tuned so that 0.98 really means right 98 times in 100
Sev1, Sev2	Highest defect severities: correctness or outage, and major with a workaround
G0 to G4	Per-tenant migration gates, from eligibility to hypercare exit
P1 to P6	Programme gates on the 12-month roadmap
Rule pack	Signed, effective-dated bundle of statutory settings, form mappings and tests
Deviation register	Signed list of intended differences from legacy behavior